

## Probability Measure

Instructor: Vishnu Boddeti

Scribe: Atharva Kirkole

## 1 Introduction

This lecture dives into probability measures, their mathematical foundations, and key examples.

### 1.1 Basic Definitions

**Definition 1 (Probability Measure)** A probability measure involves:

- An abstract space  $\Omega$  (the sample space)
- A corresponding  $\sigma$ -algebra  $\mathcal{A}$  containing measurable events

The  $\sigma$ -algebra  $\mathcal{A}$  must satisfy:

- Closure under complements: If  $A \in \mathcal{A}$ , then  $A^c \in \mathcal{A}$
- Closure under countable unions: If  $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ , then  $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$

**Definition 2 (Measure)** A measure  $\mu$  is a function  $\mu : \mathcal{A} \rightarrow [0, \infty]$  that is countably additive. For any sequence  $\{A_i\}_{i=1}^{\infty}$  of pairwise disjoint sets:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

**Definition 3 (Probability Space)** The triple  $(\Omega, \mathcal{A}, P)$  constitutes a probability space when:

- $P(\Omega) = 1$
- $P : \mathcal{A} \rightarrow [0, 1]$  is a probability measure

## 2 Examples of Probability Measures

### 2.1 Discrete Probability Measures

Single Die:

- Sample space:  $\Omega = \{1, 2, 3, 4, 5, 6\}$

- $\sigma$ -algebra:  $\mathcal{A} = \mathcal{P}(\Omega)$
- Probability measure:  $P(\{i\}) = \frac{1}{6}$  for each  $i \in \Omega$

Two Dice:

- Sample space:  $\Omega = \{(i, j) : 1 \leq i, j \leq 6\}$
- $\sigma$ -algebra:  $\mathcal{A} = \mathcal{P}(\Omega)$
- Probability measure:  $P(\{(i, j)\}) = \frac{1}{36}$  for each  $(i, j) \in \Omega$

## 2.2 Continuous Probability Measures

Normal Distribution

- Sample space:  $\Omega = \mathbb{R}$
- $\sigma$ -algebra: Borel  $\sigma$ -algebra  $\mathcal{B}(\mathbb{R})$
- Probability density function:

$$f_X(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Probability measure: For  $A \in \mathcal{B}(\mathbb{R})$ ,

$$P(A) = \int_A f_X(x|\mu, \sigma) dx$$

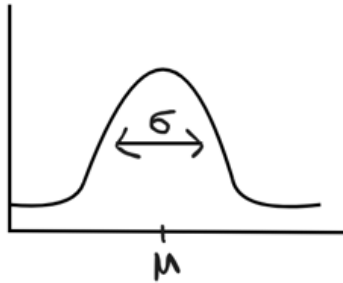


Figure 1: Fig.1 - Normal distribution

## 3 Types of Probability Measures

### 3.1 Discrete Measures

**Definition 4 (Discrete Measure)** A discrete measure on countable space  $\Omega = \{x_1, x_2, \dots\}$  with  $\mathcal{A} = \mathcal{P}(\Omega)$  satisfies:

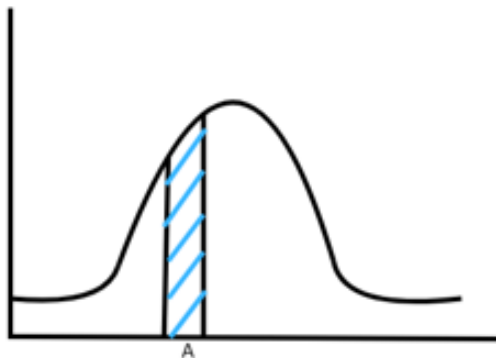


Figure 2: Fig.2 - Normal distribution Integration

- $P(\{x_i\}) = p_i$  with  $0 \leq p_i \leq 1$
- $\sum_{i=1}^{\infty} p_i = 1$

For any  $A \in \mathcal{A}$ :

$$P(A) = \sum_{\{i|x_i \in A\}} p_i$$

Coin Toss:  $\Omega = \{H, T\}$  with  $P(\{H\}) = p$  and  $P(\{T\}) = 1 - p$

### 3.2 Dirac Measures

**Definition 5 (Dirac Measure)** For  $x \in \mathbb{R}$ , the Dirac measure  $\delta_x$  on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$  is:

$$\delta_x(A) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{otherwise} \end{cases}$$

**Proposition 6** Any discrete measure on  $\mathbb{R}$  can be expressed as a sum of Dirac measures:

$$P = \sum_{i=1}^n p_i \delta_{x_i}$$

The die measure from Example 2.1 can be written as:

$$P = \frac{1}{6}(\delta_1 + \delta_2 + \cdots + \delta_6)$$

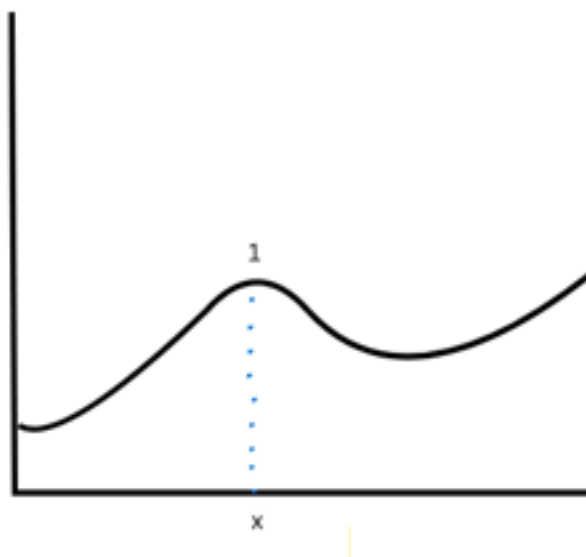


Figure 3: Fig.3 - Dirac Measure Figure

## 4 Probability Measures with Density Functions

**Definition 7 (Measure with Density)** In the context of the measure space  $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$ , where  $\lambda$  is the Lebesgue measure, a measurable function  $f : [0, \infty]$  satisfying:

$$\int_{\mathbb{R}} d\lambda = 1 \quad \text{which is} \quad \int_A f(x) dx = 1$$

defines a probability measure  $\gamma$  on  $\mathbb{R}^n$  via:

$$\gamma(A) = \int_A f(x) dx \quad \text{for all } A \in \mathcal{A}$$

$\gamma$  is the probability measure on  $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$  with density  $f$ . We denote this relationship as  $\gamma = f\lambda$ .

**Remark 8** Not all probability measures on  $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$  admit density functions. For example:

- The Dirac measure  $\delta_x$  cannot be expressed as  $\delta_x = f\lambda$  for any  $f$
- Discrete measures like  $\sum_{i=1}^n p_i \delta_{x_i}$  lack density functions

只要  $\lambda$  认为某个集合  $A$  的大小是 0 ( $\lambda(A)=0$ ) , 那么  $\mu$  必须也认为这个集合没有质量 ( $\mu(A)=0$ ) 。

## 5 Absolute Continuity of Measures

**Definition 9 (Absolute Continuity)** A probability measure  $\gamma$  on  $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$  is absolutely continuous with respect to  $\mu$  ( $\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n)$ ) if for every  $\mu$ -null set is also a  $\gamma$ -null set :  $\forall B \in \mathcal{B}(\mathbb{R}^n) :$  if  $\mu(B) = 0$  then  $\gamma(B) = 0$

$$\mu(A) = 0 \implies \gamma(A) = 0 \quad \forall A \in \mathcal{B}(\mathbb{R}^n)$$

Notation;  $\gamma \ll \mu$

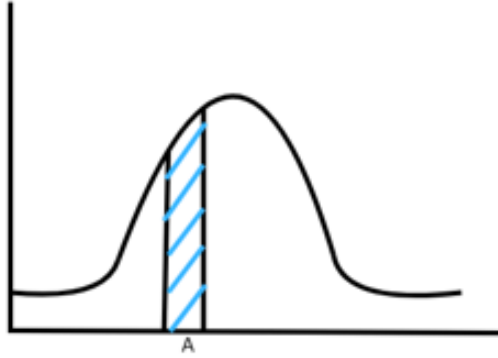


Figure 4: Fig.4 - Integral of continuous distribution

## 6 Radon-Nikodym Theorem

**Theorem 10 (Radon-Nikodym)** For  $\sigma$ -finite measures  $\mu$  and  $\gamma$  on  $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ , the following are equivalent:

1.  $\gamma$  has a density with respect to  $\mu$  (i.e.,  $\exists f$  measurable with  $\gamma(A) = \int_A f d\mu$ )
2.  $\gamma \ll \mu$  :  $\gamma$  is absolutely continuous with respect to  $\mu$

If  $\gamma \ll \mu$ , Then  $\exists \phi$  such that,  $\gamma(A) = \int_A \phi d\mu$  Then,  $\phi$  exists and is unique.

### 6.1 Proof:

#### Step 1: Define the Set of Dominating Functions

Define the set  $G$  of all measurable functions  $g : X \rightarrow [0, \infty)$  satisfying

$$\int_A g d\mu \leq \gamma(A) \quad \forall A \in \mathcal{B}.$$

- $G$  is non-empty because the zero function  $g \equiv 0$  belongs to  $G$ .
- If  $g_1, g_2 \in G$ , then  $\max(g_1, g_2) \in G$  since for any  $A$ , we can split  $A$  into  $A_1 = \{g_1 \geq g_2\}$  and  $A_2 = \{g_1 < g_2\}$ , leading to

$$\int_A \max(g_1, g_2) d\mu \leq \gamma(A).$$

#### Step 2: Define the Supremum and Construct a Maximizing Sequence

Define

$$X = \sup_{g \in G} \int_X g d\mu.$$

Since  $\gamma$  is  $\sigma$ -finite, we have  $X < \infty$ . Choose a sequence  $\{g_n\} \subset G$  such that

$$\lim_{n \rightarrow \infty} \int_X g_n d\mu = X.$$

Define the increasing sequence

$$f_n = \max(g_1, \dots, g_n),$$

which satisfies  $f_n \in G$  and  $\lim_{n \rightarrow \infty} \int_X f_n d\mu = X$  by construction.

### Step 3: Construct the Density Function

Define

$$f = \sup_n g_n.$$

By the monotone convergence theorem,

$$\int_A f d\mu = \sup_n \int_A g_n d\mu \leq \gamma(A) \quad \forall A \in \mathcal{B}.$$

However, since  $f_n \in G$  and  $\int_X f_n d\mu$  approaches the supremum  $X$ , we conclude

$$\int_A f d\mu = \gamma(A),$$

ensuring  $f$  is the required Radon–Nikodym derivative.

### Step 4: Uniqueness

Suppose there exist two such functions  $f_1$  and  $f_2$ . Then

$$\int_A (f_1 - f_2) d\mu = 0 \quad \forall A \in \mathcal{B}.$$

Let  $A = \{f_1 > f_2\}$ . Then  $\mu(A) = 0$  since  $\int_A f_1 d\mu = \int_A f_2 d\mu$ . Similarly,  $\mu(\{f_2 > f_1\}) = 0$ , implying  $f_1 = f_2$ .

Thus, the function  $f$  is unique up to sets of measure zero, completing the proof.

## 7 Bertrand paradox

Probability that a random chord exceeds  $\sqrt{3}$  Consider the unit circle and the length of a random chord. The probability that the chord length  $L > \sqrt{3}$  depends on how we define "random":

1. **Uniform endpoint method (Probability = 1/3):** Fix one endpoint and choose the other uniformly. The chord length exceeds  $\sqrt{3}$  when the angle  $\theta \in (2\pi/3, 4\pi/3)$ .
2. **Uniform radius method (Probability = 1/2):** Choose a random radius and a point on it uniformly for the chord. The condition holds when the distance  $d < 1/2$  from center.
3. **Uniform midpoint method (Probability = 1/4):** Choose the midpoint uniformly in the disk. The condition holds when the midpoint lies in a circle of radius  $1/2$ .

这反映了概率论的一个基本原则：概率空间  $(\Omega, \mathcal{F}, P)$  必须明确定义。

不同的随机实验对应不同的  $\sigma$ -代数和概率测度。

在应用中，“无差别原则” (principle of indifference) 必须小心使用，否则会导致矛盾。

We calculate each case:

1. **Endpoint method:**

$$P(L > \sqrt{3}) = P(\cos \theta < -1/2) = \frac{4\pi/3 - 2\pi/3}{2\pi} = \frac{1}{3}$$

2. **Radius method:** The length condition becomes  $2\sqrt{1-d^2} > \sqrt{3} \Rightarrow d < 1/2$ . Thus:

$$P = \frac{1/2}{1} = \frac{1}{2}$$

3. **Midpoint method:** The area where  $L > \sqrt{3}$  is a circle of radius  $1/2$ :

$$P = \frac{\pi(1/2)^2}{\pi(1)^2} = \frac{1}{4}$$

## Discussion

This example illustrates the **Bertrand paradox**, showing how probability results depend critically on the the selected defined sample space. The different probabilities  $(1/2, 1/3, 1/4)$  correspond to different natural  $\sigma$ -algebras and measures on the space of chords.

## 8 Applications in Artificial Intelligence

Probability measures play a crucial role in AI, particularly in probabilistic modeling, Bayesian inference, and deep learning. Many AI systems rely on probability distributions to model uncertainty and make informed decisions.

**Fact: Gaussian Processes in Machine Learning:** Gaussian Processes (GPs) leverage probability measures to define distributions over functions. This enables AI models to perform regression and classification tasks with built-in uncertainty. GPs are widely used in Bayesian optimization, reinforcement learning, and time-series forecasting.